

KALAT, Pakistan

WMO: 416960

Lat: 29.033N

Lon: 66.583E

Elev: 2017

StdP: 79.33

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.9	-6.0	-18.6	0.9	3.0	-16.7	1.1	1.9	4.7	5.4	3.3	7.2	0.1	N/A	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.2	33.7	16.4	32.8	15.8	31.8	15.2	20.0	28.7	18.3	27.1	17.3	26.8	0.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	16.3	23.9	15.5	14.1	23.5	13.7	12.5	23.7	68.0	28.7	61.3	27.0	57.4	26.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 3.4	2.5	2.1	-11.0	35.4	1.0	1.2	-11.7	36.2	-12.3	36.9	-12.8	37.6	-13.5	38.5		
(5)			-12.6	20.2	0.7	1.6	-13.1	21.4	-13.5	22.3	-14.0	23.2	-14.5	24.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	3.2	5.2	10.2	15.4	19.6	23.3	24.9	23.7	19.5	13.7	8.1	4.2		
	DBStd	8.17	3.03	4.00	3.36	3.48	2.90	2.35	1.65	1.82	2.61	2.87	2.94	2.91		
	HDD10.0	644	211	138	39	3	0	0	0	0	0	3	72	179		
	HDD18.3	2118	469	367	253	99	21	1	0	0	18	146	308	437		
	CDD10.0	2214	0	4	44	165	298	399	463	424	285	117	14	1		
	CDD18.3	646	0	0	0	11	60	149	204	166	53	2	0	0		
	CDH23.3	8241	0	0	9	187	880	1871	2436	1892	838	126	1	0		
	CDH26.7	3284	0	0	1	27	263	809	1142	784	238	20	0	0		
(14) Wind	WSAvg	0.4	0.4	0.6	0.4	0.4	0.5	0.4	0.4	0.4	0.3	0.4	0.4	0.4		
(15) Precipitation	PrecAvg	96	28	24	23	8	3	5	12	6	1	1	4	16		
	PrecMax	307	130	135	126	46	38	72	75	60	36	25	72	123		
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	90	37	30	37	12	8	15	19	12	6	5	14	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.6	20.1	24.7	29.3	32.1	34.4	34.6	34.7	33.9	30.9	21.9	18.5		
		MCWB	5.5	8.4	10.3	12.5	13.9	14.6	15.9	18.8	23.3	18.0	8.5	6.8		
	2%	DB	14.4	17.4	22.3	27.1	30.9	33.2	33.7	33.0	30.7	26.7	20.2	15.6		
		MCWB	5.0	7.2	9.9	12.5	13.6	14.4	15.7	17.6	16.2	11.5	8.2	5.9		
	5%	DB	12.6	15.5	20.3	25.4	29.6	32.1	32.8	31.8	29.2	24.7	18.3	14.1		
		MCWB	4.4	6.6	8.9	12.3	13.2	14.3	15.6	15.6	14.2	11.0	7.5	5.1		
	10%	DB	10.8	13.5	18.3	23.8	28.1	30.9	31.9	30.7	28.0	22.6	16.6	12.3		
		MCWB	3.6	5.5	8.1	11.7	12.7	14.1	15.3	14.9	13.3	10.0	6.8	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.6	9.8	12.7	17.7	17.0	19.2	19.8	25.7	24.9	18.7	11.0	8.8		
		MCDB	11.9	17.7	20.1	24.4	28.2	25.2	28.3	32.7	31.7	28.8	17.9	13.5		
	2%	WB	6.3	8.3	10.8	15.1	15.1	17.9	18.6	19.7	20.8	13.1	9.6	7.0		
		MCDB	11.8	15.3	19.6	22.4	27.7	26.7	28.1	29.3	28.5	23.5	17.1	13.1		
	5%	WB	5.4	7.2	9.7	13.2	14.0	17.0	17.7	18.2	16.1	11.6	8.4	5.9		
		MCDB	11.0	14.2	18.6	22.6	27.2	26.7	27.8	26.4	24.1	22.3	16.8	12.5		
	10%	WB	4.4	6.0	8.7	12.1	13.3	15.7	16.9	16.9	14.3	10.5	7.4	4.9		
		MCDB	10.1	12.8	17.0	22.7	26.5	27.3	27.7	26.1	25.8	21.5	15.7	11.6		
(35) Mean Daily Temperature Range	MDBR		13.9	12.9	13.5	14.1	15.3	15.2	14.2	14.5	16.7	17.1	15.8	14.8		
	5% DB	MCDBR	16.6	15.9	16.9	15.6	17.1	16.1	15.0	15.6	17.7	18.9	17.5	17.3		
		MCWBR	10.0	8.5	8.1	7.3	6.2	5.8	5.5	6.1	7.9	8.1	8.7	9.4		
	5% WB	MCDBR	14.4	13.7	14.8	14.2	15.5	13.9	13.7	13.0	15.9	17.1	15.5	14.6		
		MCWBR	10.2	8.8	9.1	9.1	6.9	7.0	6.5	7.5	10.2	8.4	8.7	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.334	0.372	0.425	0.433	0.466	0.572	0.506	0.385	0.326	0.309	0.278		
	taud		2.401	2.264	2.126	1.996	1.975	1.898	1.696	1.832	2.123	2.263	2.322	2.431		
	Ebn at Noon		956	931	913	872	862	829	743	791	887	926	919	952		
	Edh at Noon		99	124	152	179	184	198	241	209	150	122	106	91		
(44) All-Sky Solar Radiation	RadAvg		3.87	4.55	5.88	6.78	7.53	7.78	7.05	6.79	6.56	5.74	4.50	3.81		
	RadStd		0.26	0.40	0.36	0.37	0.30	0.32	0.38	0.37	0.23	0.15	0.20	0.21		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page